Differentiate each function with respect to x.

1) $y = \sin^{-1} 4x^4$

A) $\frac{dy}{dx} = 16x^3 \cos 4x^4$

B) $\frac{dy}{dx} = \frac{16x^3}{1 - 16x^8}$

C) $\frac{dy}{dx} = \frac{16x^3}{\sqrt{1 - 4x^4}}$

D) $\frac{dy}{dx} = \frac{16x^3}{\sqrt{1 - 16x^8}}$

$u =$

$u' =$

$u^2 =$

2) $y = \tan^{-1} 3x^3$

A) $\frac{dy}{dx} = \frac{9x^2}{\sqrt{9x^6 + 1}}$

B) $\frac{dy}{dx} = \frac{9x^2}{9x^6 + 1}$

C) $\frac{dy}{dx} = \frac{9x^2}{3x^3 + 1}$

D) $\frac{dy}{dx} = 9x^2 \sec^2 3x^3$

$u =$

$u' =$

$u^2 =$

Differentiate each function with respect to *x*.

3) $y = \cot^{-1} x^5$

A) $\frac{dy}{dx} = -\frac{5x^4}{x^5 + 1}$

B) $\frac{dy}{dx} = -\frac{5x^4}{\sqrt{x^{10} + 1}}$

C) $\frac{dy}{dx} = -5x^4 \sin^2 x^5$

D) $\frac{dy}{dx} = -\frac{5x^4}{x^{10} + 1}$

$u =$

$u' =$

$u^2 =$

4) $y = \cos^{-1} 4x^5$

A) $\frac{dy}{dx} = -\frac{20x^4}{1 - 16x^{10}}$

B) $\frac{dy}{dx} = -\frac{20x^4}{\sqrt{1 - 16x^{10}}}$

C) $\frac{dy}{dx} = -20x^4 \sin 4x^5$

D) $\frac{dy}{dx} = -\frac{20x^4}{\sqrt{1 - 4x^5}}$

$u =$

$u' =$

$u^2 =$